

nD: 4, H: 1, H+: 1

term0

$$+ \left(\frac{1}{24} a_t^2 - \frac{1}{4} a_t \hat{\mu} + \frac{1}{4} \ln \frac{M_s^2}{\mu_M^2} a_t \hat{\mu} - \frac{1}{2} \frac{a_t \hat{\mu}}{\varepsilon} \right) M_s^{-2} \quad (1)$$

$$\varepsilon_{ij} \varepsilon^{\alpha_2 \alpha_1} \varepsilon^{\alpha_4 \alpha_3} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} \varepsilon_{\dot{\alpha}_4 \dot{\alpha}_3} D_{\dot{\alpha}_1}^{\dot{\alpha}_1} D_{\dot{\alpha}_2}^{\dot{\alpha}_2} H_i D_{\dot{\alpha}_3}^{\dot{\alpha}_3} D_{\dot{\alpha}_4}^{\dot{\alpha}_4} (H^\dagger)_j$$

term3

$$- \left(\frac{1}{4} a_t \hat{\mu} - \frac{1}{4} \ln \frac{M_s^2}{\mu_M^2} a_t \hat{\mu} + \frac{1}{2} \frac{a_t \hat{\mu}}{\varepsilon} \right) M_s^{-2} \quad (2)$$

$$\varepsilon_{ij} \varepsilon^{\alpha_1 \alpha_4} \varepsilon^{\alpha_3 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_4} \varepsilon_{\dot{\alpha}_3 \dot{\alpha}_2} D_{\dot{\alpha}_1}^{\dot{\alpha}_1} H_i D_{\dot{\alpha}_2}^{\dot{\alpha}_2} D_{\dot{\alpha}_3}^{\dot{\alpha}_3} D_{\dot{\alpha}_4}^{\dot{\alpha}_4} (H^\dagger)_j$$

term4

$$- \left(\frac{1}{4} a_t \hat{\mu} - \frac{1}{4} \ln \frac{M_s^2}{\mu_M^2} a_t \hat{\mu} + \frac{1}{2} \frac{a_t \hat{\mu}}{\varepsilon} \right) M_s^{-2} \quad (3)$$

$$\varepsilon_{ij} \varepsilon^{\alpha_3 \alpha_2} \varepsilon^{\alpha_1 \alpha_4} \varepsilon_{\dot{\alpha}_3 \dot{\alpha}_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_4} D_{\dot{\alpha}_1}^{\dot{\alpha}_1} D_{\dot{\alpha}_2}^{\dot{\alpha}_2} D_{\dot{\alpha}_3}^{\dot{\alpha}_3} H_i D_{\dot{\alpha}_4}^{\dot{\alpha}_4} (H^\dagger)_j$$

term5

$$- \left(\frac{1}{2} a_t \hat{\mu} - \frac{1}{2} \ln \frac{M_s^2}{\mu_M^2} a_t \hat{\mu} + \frac{a_t \hat{\mu}}{\varepsilon} \right) M_s^{-2} \quad (4)$$

$$\varepsilon_{ij} \varepsilon^{\alpha_2 \alpha_4} \varepsilon^{\alpha_1 \alpha_3} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_4} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_3} D_{\dot{\alpha}_1}^{\dot{\alpha}_1} D_{\dot{\alpha}_2}^{\dot{\alpha}_2} H_i D_{\dot{\alpha}_3}^{\dot{\alpha}_3} D_{\dot{\alpha}_4}^{\dot{\alpha}_4} (H^\dagger)_j$$

term6

$$- \left(\frac{1}{8} a_t \hat{\mu} - \frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t \hat{\mu} + \frac{1}{4} \frac{a_t \hat{\mu}}{\varepsilon} \right) M_s^{-2} \quad (5)$$

$$\varepsilon_{ij} \varepsilon^{\alpha_4 \alpha_3} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_4 \dot{\alpha}_3} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} D_{\dot{\alpha}_1}^{\dot{\alpha}_1} D_{\dot{\alpha}_2}^{\dot{\alpha}_2} D_{\dot{\alpha}_3}^{\dot{\alpha}_3} D_{\dot{\alpha}_4}^{\dot{\alpha}_4} H_i (H^\dagger)_j$$

term7

$$- \left(\frac{1}{4} a_t \hat{\mu} - \frac{1}{4} \ln \frac{M_s^2}{\mu_M^2} a_t \hat{\mu} + \frac{1}{2} \frac{a_t \hat{\mu}}{\varepsilon} \right) M_s^{-2} \quad (6)$$

$$\varepsilon_{ij} \varepsilon^{\alpha_3 \alpha_4} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_3 \dot{\alpha}_4} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} D_{\dot{\alpha}_1}^{\dot{\alpha}_1} D_{\dot{\alpha}_2}^{\dot{\alpha}_2} D_{\dot{\alpha}_3}^{\dot{\alpha}_3} H_i D_{\dot{\alpha}_4}^{\dot{\alpha}_4} (H^\dagger)_j$$

term8

$$- \left(\frac{1}{4} a_t \hat{\mu} - \frac{1}{4} \ln \frac{M_s^2}{\mu_M^2} a_t \hat{\mu} + \frac{1}{2} \frac{a_t \hat{\mu}}{\varepsilon} \right) M_s^{-2} \quad (7)$$

$$\varepsilon_{ij} \varepsilon^{\alpha_1 \alpha_2} \varepsilon^{\alpha_4 \alpha_3} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \varepsilon_{\dot{\alpha}_4 \dot{\alpha}_3} D_{\dot{\alpha}_1}^{\dot{\alpha}_1} H_i D_{\dot{\alpha}_2}^{\dot{\alpha}_2} D_{\dot{\alpha}_3}^{\dot{\alpha}_3} D_{\dot{\alpha}_4}^{\dot{\alpha}_4} (H^\dagger)_j$$

term10

$$- \left(\frac{1}{8} a_t \hat{\mu} - \frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t \hat{\mu} + \frac{1}{4} \frac{a_t \hat{\mu}}{\varepsilon} \right) M_s^{-2} \quad (8)$$

$$\varepsilon_{ij} \varepsilon^{\alpha_4 \alpha_3} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_4 \dot{\alpha}_3} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} H_i D_{\dot{\alpha}_1}^{\dot{\alpha}_1} D_{\dot{\alpha}_2}^{\dot{\alpha}_2} D_{\dot{\alpha}_3}^{\dot{\alpha}_3} D_{\dot{\alpha}_4}^{\dot{\alpha}_4} (H^\dagger)_j$$

nD: 2, H: 1, H+: 1

term1

$$- \left(\frac{1}{4} a_t^2 + \frac{1}{2} a_t \hat{\mu} - \frac{1}{2} \ln \frac{M_s^2}{\mu_M^2} a_t \hat{\mu} - \frac{1}{2} \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 + \frac{5}{4} a_t^2 \hat{\mu}_h^2 + \frac{a_t \hat{\mu}}{\varepsilon} + \frac{a_t^2 \hat{\mu}_h^2}{\varepsilon} \right) \quad (9)$$

$$\varepsilon_{ij} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} D_{\dot{\alpha}_1}^{\dot{\alpha}_1} H_i D_{\dot{\alpha}_2}^{\dot{\alpha}_2} (H^\dagger)_j$$

term2

$$-\left(\frac{1}{4}a_t\hat{\mu}-\frac{1}{4}\ln\frac{M_s^2}{\mu_M^2}a_t\hat{\mu}+\frac{1}{2}\frac{a_t\hat{\mu}}{\varepsilon}\right) \varepsilon_{ij}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_1}D_{\alpha_1}^{\dot{\alpha}_1}D_{\alpha_2}^{\dot{\alpha}_2}H_i(H^\dagger)_j \quad (10)$$

term9

$$-\left(\frac{1}{4}a_t\hat{\mu}-\frac{1}{4}\ln\frac{M_s^2}{\mu_M^2}a_t\hat{\mu}+\frac{1}{2}\frac{a_t\hat{\mu}}{\varepsilon}\right) \varepsilon_{ij}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_1}H_iD_{\alpha_1}^{\dot{\alpha}_1}D_{\alpha_2}^{\dot{\alpha}_2}(H^\dagger)_j \quad (11)$$

nD: 0, H: 1, H+: 1

term11

$$+\left(a_t^2+a_t^2\hat{\mu}_h^2+a_t^2\hat{\mu}_h^4-\ln\frac{M_s^2}{\mu_M^2}a_t^2-\ln\frac{M_s^2}{\mu_M^2}a_t^2\hat{\mu}_h^2-\ln\frac{M_s^2}{\mu_M^2}a_t^2\hat{\mu}_h^4-\frac{1}{2}a_t\hat{\mu}+\frac{1}{2}\ln\frac{M_s^2}{\mu_M^2}a_t\hat{\mu}+\frac{1}{2}\kappa-\frac{1}{2}\kappa\ln\frac{M_s^2}{\mu_M^2}+2\frac{a_t^2}{\varepsilon}+2\frac{a_t^2\hat{\mu}_h^2}{\varepsilon}+2\frac{a_t^2\hat{\mu}_h^4}{\varepsilon}-\frac{a_t\hat{\mu}}{\varepsilon}+\frac{\kappa}{\varepsilon}\right)M_s^2 \varepsilon_{ij}H_i(H^\dagger)_j \quad (12)$$

nD: 2, BL: 1, H: 1, H+: 1

term12

$$-\left(\frac{1}{96}g_1\ln\frac{M_s^2}{\mu_M^2}a_t^2-\frac{1}{48}\frac{g_1a_t^2}{\varepsilon}-\frac{17}{576}g_1a_t^2\right)M_s^{-2} \varepsilon_{ij}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}\varepsilon^{\alpha_1\alpha_3}\varepsilon^{\alpha_2\alpha_4}(B_L)_{\alpha_1\alpha_2}D_{\alpha_3}^{\dot{\alpha}_1}H_iD_{\alpha_4}^{\dot{\alpha}_2}(H^\dagger)_j \quad (13)$$

term12

$$-\left(\frac{1}{96}g_1\ln\frac{M_s^2}{\mu_M^2}a_t^2-\frac{1}{48}\frac{g_1a_t^2}{\varepsilon}-\frac{17}{576}g_1a_t^2\right)M_s^{-2} \varepsilon_{ij}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}\varepsilon^{\alpha_1\alpha_4}\varepsilon^{\alpha_2\alpha_3}(B_L)_{\alpha_1\alpha_2}D_{\alpha_3}^{\dot{\alpha}_1}H_iD_{\alpha_4}^{\dot{\alpha}_2}(H^\dagger)_j \quad (14)$$

term13

$$-\left(\frac{1}{96}g_1\ln\frac{M_s^2}{\mu_M^2}a_t^2-\frac{1}{48}\frac{g_1a_t^2}{\varepsilon}-\frac{17}{576}g_1a_t^2\right)M_s^{-2} \varepsilon_{ij}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}\varepsilon^{\alpha_1\alpha_3}\varepsilon^{\alpha_2\alpha_4}(B_L)_{\alpha_1\alpha_2}D_{\alpha_3}^{\dot{\alpha}_1}H_iD_{\alpha_4}^{\dot{\alpha}_2}(H^\dagger)_j \quad (15)$$

term13

$$-\left(\frac{1}{96}g_1\ln\frac{M_s^2}{\mu_M^2}a_t^2-\frac{1}{48}\frac{g_1a_t^2}{\varepsilon}-\frac{17}{576}g_1a_t^2\right)M_s^{-2} \varepsilon_{ij}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}\varepsilon^{\alpha_1\alpha_4}\varepsilon^{\alpha_2\alpha_3}(B_L)_{\alpha_1\alpha_2}D_{\alpha_3}^{\dot{\alpha}_1}H_iD_{\alpha_4}^{\dot{\alpha}_2}(H^\dagger)_j \quad (16)$$

nD: 2, H: 1, H+: 1, BL+: 1

term12

$$+\left(\frac{1}{96}g_1\ln\frac{M_s^2}{\mu_M^2}a_t^2-\frac{1}{48}\frac{g_1a_t^2}{\varepsilon}-\frac{17}{576}g_1a_t^2\right)M_s^{-2} \varepsilon_{ij}\varepsilon^{\alpha_1\alpha_2}\varepsilon_{\dot{\alpha}_3\dot{\alpha}_1}\varepsilon_{\dot{\alpha}_4\dot{\alpha}_2}D_{\alpha_1}^{\dot{\alpha}_1}H_iD_{\alpha_2}^{\dot{\alpha}_2}(H^\dagger)_j(B_R)^{\dot{\alpha}_3\dot{\alpha}_4} \quad (17)$$

term12

$$+ \left(\frac{1}{96} g_1 \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{48} \frac{g_1 a_t^2}{\varepsilon} - \frac{17}{576} g_1 a_t^2 \right) M_s^{-2} \quad (18)$$

$$\varepsilon_{ij} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_3 \dot{\alpha}_2} \varepsilon_{\dot{\alpha}_4 \dot{\alpha}_1} D_{\alpha_1}^{\dot{\alpha}_1} H_i D_{\alpha_2}^{\dot{\alpha}_2} (H^\dagger)_j (B_R)^{\dot{\alpha}_3 \dot{\alpha}_4}$$

term13

$$+ \left(\frac{1}{96} g_1 \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{48} \frac{g_1 a_t^2}{\varepsilon} - \frac{17}{576} g_1 a_t^2 \right) M_s^{-2} \quad (19)$$

$$\varepsilon_{ij} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_3 \dot{\alpha}_1} \varepsilon_{\dot{\alpha}_4 \dot{\alpha}_2} D_{\alpha_1}^{\dot{\alpha}_1} H_i D_{\alpha_2}^{\dot{\alpha}_2} (H^\dagger)_j (B_R)^{\dot{\alpha}_3 \dot{\alpha}_4}$$

term13

$$+ \left(\frac{1}{96} g_1 \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{48} \frac{g_1 a_t^2}{\varepsilon} - \frac{17}{576} g_1 a_t^2 \right) M_s^{-2} \quad (20)$$

$$\varepsilon_{ij} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_3 \dot{\alpha}_2} \varepsilon_{\dot{\alpha}_4 \dot{\alpha}_1} D_{\alpha_1}^{\dot{\alpha}_1} H_i D_{\alpha_2}^{\dot{\alpha}_2} (H^\dagger)_j (B_R)^{\dot{\alpha}_3 \dot{\alpha}_4}$$

nD: 2, H: 2, H+: 2

term14

$$+ \left(a_t^3 \hat{\mu} \hat{\mu}_h^2 - a_t^4 \hat{\mu}_h^2 + \ln \frac{M_s^2}{\mu_M^2} a_t^4 \hat{\mu}_h^2 - \kappa a_t^2 - \kappa a_t^2 \hat{\mu}_h^2 - \kappa a_t^2 \hat{\mu}_h^4 + \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^2 \right. \\ + \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 + \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^4 + \frac{3}{32} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}^2 + \frac{1}{8} \lambda_\varphi a_t^2 \\ - \frac{1}{8} \lambda_\varphi \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{9}{64} g_2^2 a_t^2 - \frac{3}{16} \frac{g_2^2 a_t^2}{\varepsilon} - \frac{1}{4} a_t^2 \hat{\mu}^2 - \frac{1}{4} \kappa \ln \frac{M_s^2}{\mu_M^2} a_t \hat{\mu} - \frac{1}{4} \frac{a_t^2 \hat{\mu}^2}{\varepsilon} \\ + \frac{1}{4} \frac{\lambda_\varphi a_t^2}{\varepsilon} + \frac{1}{2} a_t^3 \hat{\mu} - \frac{1}{2} a_t^4 - \frac{1}{2} \ln \frac{M_s^2}{\mu_M^2} a_t^3 \hat{\mu} \hat{\mu}_h^2 + \frac{1}{2} \ln \frac{M_s^2}{\mu_M^2} a_t^4 + \frac{1}{2} \kappa a_t \hat{\mu} \\ + \frac{1}{2} \frac{\kappa a_t \hat{\mu}}{\varepsilon} - 2 \frac{a_t^4 \hat{\mu}_h^2}{\varepsilon} - 2 \frac{\kappa a_t^2}{\varepsilon} - 2 \frac{\kappa a_t^2 \hat{\mu}_h^2}{\varepsilon} - 2 \frac{\kappa a_t^2 \hat{\mu}_h^4}{\varepsilon} + \frac{a_t^3 \hat{\mu} \hat{\mu}_h^2}{\varepsilon} - \frac{a_t^4}{\varepsilon} \Big) M_s^{-2} \quad (21)$$

$$\varepsilon_{il} \varepsilon_{jk} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} D_{\alpha_1}^{\dot{\alpha}_1} D_{\alpha_2}^{\dot{\alpha}_2} H_i H_j (H^\dagger)_k (H^\dagger)_l$$

term15

$$- \left(\frac{3}{32} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}^2 - \frac{1}{8} \lambda_\varphi a_t^2 + \frac{1}{8} \lambda_\varphi \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{9}{64} g_2^2 a_t^2 - \frac{3}{16} \frac{g_2^2 a_t^2}{\varepsilon} \right. \\ + \frac{1}{4} a_t^2 \hat{\mu}^2 - \frac{1}{4} \kappa a_t \hat{\mu} + \frac{1}{4} \kappa \ln \frac{M_s^2}{\mu_M^2} a_t \hat{\mu} + \frac{1}{4} \frac{a_t^2 \hat{\mu}^2}{\varepsilon} - \frac{1}{4} \frac{\lambda_\varphi a_t^2}{\varepsilon} - \frac{1}{2} \frac{\kappa a_t \hat{\mu}}{\varepsilon} \Big) M_s^{-2} \quad (22)$$

$$\varepsilon_{ik} \varepsilon_{jl} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} D_{\alpha_1}^{\dot{\alpha}_1} D_{\alpha_2}^{\dot{\alpha}_2} H_i H_j (H^\dagger)_k (H^\dagger)_l$$

term16

$$\begin{aligned}
& + \left(a_t^3 \hat{\mu} \hat{\mu}_h^2 - a_t^4 \hat{\mu}_h^2 + \ln \frac{M_s^2}{\mu_M^2} a_t^4 \hat{\mu}_h^2 - \kappa a_t^2 - \kappa a_t^2 \hat{\mu}_h^2 - \kappa a_t^2 \hat{\mu}_h^4 + \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^2 \right. \\
& + \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 + \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^4 + \frac{3}{32} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}^2 + \frac{1}{8} \lambda_\varphi a_t^2 \\
& - \frac{1}{8} \lambda_\varphi \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{9}{64} g_2^2 a_t^2 - \frac{3}{16} \frac{g_2^2 a_t^2}{\varepsilon} - \frac{1}{4} a_t^2 \hat{\mu}^2 - \frac{1}{4} \kappa \ln \frac{M_s^2}{\mu_M^2} a_t \hat{\mu} - \frac{1}{4} \frac{a_t^2 \hat{\mu}^2}{\varepsilon} \\
& + \frac{1}{4} \frac{\lambda_\varphi a_t^2}{\varepsilon} + \frac{1}{2} a_t^3 \hat{\mu} - \frac{1}{2} a_t^4 - \frac{1}{2} \ln \frac{M_s^2}{\mu_M^2} a_t^3 \hat{\mu} \hat{\mu}_h^2 + \frac{1}{2} \ln \frac{M_s^2}{\mu_M^2} a_t^4 + \frac{1}{2} \kappa a_t \hat{\mu} \\
& \left. + \frac{1}{2} \frac{\kappa a_t \hat{\mu}}{\varepsilon} - 2 \frac{a_t^4 \hat{\mu}_h^2}{\varepsilon} - 2 \frac{\kappa a_t^2}{\varepsilon} - 2 \frac{\kappa a_t^2 \hat{\mu}_h^2}{\varepsilon} - 2 \frac{\kappa a_t^2 \hat{\mu}_h^4}{\varepsilon} + \frac{a_t^3 \hat{\mu} \hat{\mu}_h^2}{\varepsilon} - \frac{a_t^4}{\varepsilon} \right) M_s^{-2} \\
& \varepsilon_{il} \varepsilon_{jk} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} H_i H_j D_{\alpha_1}^{\dot{\alpha}_1} D_{\alpha_2}^{\dot{\alpha}_2} (H^\dagger)_k (H^\dagger)_l
\end{aligned} \tag{23}$$

term17

$$\begin{aligned}
& - \left(\frac{3}{32} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}^2 - \frac{1}{8} \lambda_\varphi a_t^2 + \frac{1}{8} \lambda_\varphi \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{9}{64} g_2^2 a_t^2 - \frac{3}{16} \frac{g_2^2 a_t^2}{\varepsilon} \right. \\
& \left. + \frac{1}{4} a_t^2 \hat{\mu}^2 - \frac{1}{4} \kappa a_t \hat{\mu} + \frac{1}{4} \kappa \ln \frac{M_s^2}{\mu_M^2} a_t \hat{\mu} + \frac{1}{4} \frac{a_t^2 \hat{\mu}^2}{\varepsilon} - \frac{1}{4} \frac{\lambda_\varphi a_t^2}{\varepsilon} - \frac{1}{2} \frac{\kappa a_t \hat{\mu}}{\varepsilon} \right) M_s^{-2} \\
& \varepsilon_{jk} \varepsilon_{il} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} H_i H_j D_{\alpha_1}^{\dot{\alpha}_1} D_{\alpha_2}^{\dot{\alpha}_2} (H^\dagger)_k (H^\dagger)_l
\end{aligned} \tag{24}$$

term18

$$\begin{aligned}
& - \left(a_t^2 \hat{\mu}^2 + \ln \frac{M_s^2}{\mu_M^2} a_t^3 \hat{\mu} \hat{\mu}_h^2 + \kappa \ln \frac{M_s^2}{\mu_M^2} a_t \hat{\mu} - \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 + \frac{1}{24} d g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 \right. \\
& - \frac{11}{144} d g_2^2 a_t^2 - \frac{1}{12} \frac{d g_2^2 a_t^2}{\varepsilon} - \frac{1}{8} \varepsilon \ln \frac{M_s^2}{\mu_M^2} a_t^3 \hat{\mu} + \frac{1}{8} \varepsilon \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{1}{4} \varepsilon a_t^3 \hat{\mu} \\
& - \frac{1}{4} \varepsilon \kappa a_t^2 - \frac{29}{72} g_2^2 a_t^2 - \frac{1}{2} \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}^2 - \frac{1}{2} \ln \frac{M_s^2}{\mu_M^2} a_t^4 - \frac{1}{2} \kappa a_t^2 \hat{\mu}_h^2 - \frac{1}{2} \lambda_\varphi a_t^2 \\
& + \frac{1}{2} \lambda_\varphi \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{2} g_2^2 a_t^2 \hat{\mu}_h^2 + \frac{1}{2} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 - \frac{1}{2} \lambda_h \ln \frac{M_s^2}{\mu_M^2} a_t^2 \\
& + \frac{1}{2} \varepsilon \ln \frac{M_s^2}{\mu_M^2} a_t^4 + \frac{7}{12} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{4} a_t^3 \hat{\mu} - \frac{3}{4} \varepsilon a_t^4 - \frac{7}{6} \frac{g_2^2 a_t^2}{\varepsilon} - \frac{5}{4} a_t^4 + \frac{5}{4} \kappa a_t^2 \\
& - \frac{3}{2} \kappa a_t \hat{\mu} + \frac{9}{4} \lambda_h a_t^2 - 2 a_t^3 \hat{\mu} \hat{\mu}_h^2 + 2 a_t^4 \hat{\mu}_h^2 - 2 \ln \frac{M_s^2}{\mu_M^2} a_t^4 \hat{\mu}_h^2 + 2 \kappa a_t^2 \hat{\mu}_h^4 \\
& - 2 \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^2 - 2 \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^4 - 2 \frac{a_t^3 \hat{\mu} \hat{\mu}_h^2}{\varepsilon} - 2 \frac{\kappa a_t \hat{\mu}}{\varepsilon} + 2 \frac{\kappa a_t^2 \hat{\mu}_h^2}{\varepsilon} + 4 \frac{a_t^4 \hat{\mu}_h^2}{\varepsilon} \\
& \left. + 4 \frac{\kappa a_t^2}{\varepsilon} + 4 \frac{\kappa a_t^2 \hat{\mu}_h^4}{\varepsilon} + \frac{a_t^2 \hat{\mu}^2}{\varepsilon} + \frac{a_t^4}{\varepsilon} - \frac{\lambda_\varphi a_t^2}{\varepsilon} - \frac{g_2^2 a_t^2 \hat{\mu}_h^2}{\varepsilon} + \frac{\lambda_h a_t^2}{\varepsilon} \right) M_s^{-2} \\
& \varepsilon_{ik} \varepsilon_{jl} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} D_{\alpha_1}^{\dot{\alpha}_1} H_i H_j D_{\alpha_2}^{\dot{\alpha}_2} (H^\dagger)_k (H^\dagger)_l
\end{aligned} \tag{25}$$

term19

$$\begin{aligned}
& + \left(\frac{1}{96} dg_1^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{11}{576} dg_1^2 a_t^2 - \frac{1}{48} \kappa^2 + \frac{1}{48} dg_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{48} \frac{dg_1^2 a_t^2}{\varepsilon} - \frac{11}{288} dg_2^2 a_t^2 \right. \\
& - \frac{1}{24} g_1^2 a_t^2 - \frac{1}{24} \frac{dg_2^2 a_t^2}{\varepsilon} - \frac{5}{96} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{5}{48} \frac{g_2^2 a_t^2}{\varepsilon} + \frac{1}{8} g_2^2 a_t^2 \hat{\mu}_h^2 - \frac{1}{8} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 \\
& - \frac{1}{8} g_1^2 a_t^2 \hat{\mu}_h^2 + \frac{1}{8} g_1^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{1}{8} g_1^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 - \frac{1}{6} a_t^3 \hat{\mu} - \frac{3}{16} \varepsilon \ln \frac{M_s^2}{\mu_M^2} a_t^4 + \frac{121}{576} g_2^2 a_t^2 \\
& + \frac{1}{4} \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}^2 - \frac{1}{4} \ln \frac{M_s^2}{\mu_M^2} a_t^4 + \frac{1}{4} \lambda_h \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{1}{4} \frac{g_2^2 a_t^2 \hat{\mu}_h^2}{\varepsilon} - \frac{1}{4} \frac{g_1^2 a_t^2}{\varepsilon} - \frac{1}{4} \frac{g_1^2 a_t^2 \hat{\mu}_h^2}{\varepsilon} \\
& - \frac{13}{48} a_t^2 \hat{\mu}^2 + \frac{11}{32} \varepsilon a_t^4 + \frac{5}{12} \kappa a_t^2 - \frac{1}{2} \kappa \ln \frac{M_s^2}{\mu_M^2} a_t \hat{\mu} - \frac{1}{2} \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 - \frac{1}{2} \frac{a_t^2 \hat{\mu}^2}{\varepsilon} + \frac{1}{2} \frac{a_t^4}{\varepsilon} \\
& - \frac{1}{2} \frac{\lambda_h a_t^2}{\varepsilon} + \frac{13}{24} \kappa a_t \hat{\mu} - \frac{7}{8} \lambda_h a_t^2 + \frac{7}{6} a_t^4 + \frac{5}{4} \kappa a_t^2 \hat{\mu}_h^2 + \frac{\kappa a_t \hat{\mu}}{\varepsilon} + \frac{\kappa a_t^2 \hat{\mu}_h^2}{\varepsilon} \Big) M_s^{-2} \\
& \varepsilon_{ik} \varepsilon_{jl} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} D_{\alpha_1}^{\dot{\alpha}_1} H_i D_{\alpha_2}^{\dot{\alpha}_2} H_j (H^\dagger)_k (H^\dagger)_l
\end{aligned} \tag{26}$$

term20

$$\begin{aligned}
& - \left(\kappa \ln \frac{M_s^2}{\mu_M^2} a_t \hat{\mu} + \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 - \frac{1}{48} dg_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{1}{48} dg_1^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{11}{288} dg_2^2 a_t^2 \right. \\
& - \frac{11}{288} dg_1^2 a_t^2 + \frac{1}{24} \kappa^2 + \frac{1}{24} \frac{dg_2^2 a_t^2}{\varepsilon} - \frac{1}{24} \frac{dg_1^2 a_t^2}{\varepsilon} + \frac{29}{144} g_2^2 a_t^2 - \frac{29}{144} g_1^2 a_t^2 + \frac{1}{4} g_2^2 a_t^2 \hat{\mu}_h^2 \\
& - \frac{1}{4} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 - \frac{1}{4} g_1^2 a_t^2 \hat{\mu}_h^2 + \frac{1}{4} g_1^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 - \frac{7}{24} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{7}{24} g_1^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 \\
& + \frac{1}{3} a_t^3 \hat{\mu} + \frac{3}{8} \varepsilon \ln \frac{M_s^2}{\mu_M^2} a_t^4 - \frac{1}{2} \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}^2 + \frac{1}{2} \ln \frac{M_s^2}{\mu_M^2} a_t^4 - \frac{1}{2} \lambda_h \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{1}{2} \frac{g_2^2 a_t^2 \hat{\mu}_h^2}{\varepsilon} \\
& - \frac{1}{2} \frac{g_1^2 a_t^2 \hat{\mu}_h^2}{\varepsilon} + \frac{13}{24} a_t^2 \hat{\mu}^2 + \frac{7}{12} \frac{g_2^2 a_t^2}{\varepsilon} - \frac{7}{12} \frac{g_1^2 a_t^2}{\varepsilon} - \frac{11}{16} \varepsilon a_t^4 - \frac{5}{6} \kappa a_t^2 - \frac{13}{12} \kappa a_t \hat{\mu} \\
& + \frac{9}{4} \lambda_h a_t^2 - \frac{7}{3} a_t^4 - \frac{5}{2} \kappa a_t^2 \hat{\mu}_h^2 - 2 \frac{\kappa a_t \hat{\mu}}{\varepsilon} - 2 \frac{\kappa a_t^2 \hat{\mu}_h^2}{\varepsilon} + \frac{a_t^2 \hat{\mu}^2}{\varepsilon} - \frac{a_t^4}{\varepsilon} + \frac{\lambda_h a_t^2}{\varepsilon} \Big) M_s^{-2} \\
& \varepsilon_{il} \varepsilon_{jk} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} D_{\alpha_1}^{\dot{\alpha}_1} H_i H_j D_{\alpha_2}^{\dot{\alpha}_2} (H^\dagger)_k (H^\dagger)_l
\end{aligned} \tag{27}$$

term21

$$\begin{aligned}
& - \left(\frac{1}{96} dg_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{11}{576} dg_2^2 a_t^2 - \frac{1}{48} g_1^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{48} \frac{dg_2^2 a_t^2}{\varepsilon} \right. \\
& + \frac{1}{24} \frac{g_1^2 a_t^2}{\varepsilon} + \frac{17}{288} g_1^2 a_t^2 - \frac{19}{96} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{1}{4} g_2^2 a_t^2 \hat{\mu}_h^2 - \frac{1}{4} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 \\
& + \frac{1}{4} \lambda_h a_t^2 + \frac{179}{576} g_2^2 a_t^2 + \frac{19}{48} \frac{g_2^2 a_t^2}{\varepsilon} + \frac{1}{2} \frac{g_2^2 a_t^2 \hat{\mu}_h^2}{\varepsilon} \Big) M_s^{-2} \\
& \varepsilon_{il} \varepsilon_{jk} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} D_{\alpha_1}^{\dot{\alpha}_1} H_i D_{\alpha_2}^{\dot{\alpha}_2} H_j (H^\dagger)_k (H^\dagger)_l
\end{aligned} \tag{28}$$

term23

$$\begin{aligned}
& + \left(\frac{1}{96} dg_1^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{11}{576} dg_1^2 a_t^2 - \frac{1}{48} \kappa^2 + \frac{1}{48} dg_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{48} \frac{dg_1^2 a_t^2}{\varepsilon} - \frac{11}{288} dg_2^2 a_t^2 \right. \\
& - \frac{1}{24} g_1^2 a_t^2 - \frac{1}{24} \frac{dg_2^2 a_t^2}{\varepsilon} - \frac{5}{96} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{5}{48} \frac{g_2^2 a_t^2}{\varepsilon} + \frac{1}{8} g_2^2 a_t^2 \hat{\mu}_h^2 - \frac{1}{8} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 \\
& - \frac{1}{8} g_1^2 a_t^2 \hat{\mu}_h^2 + \frac{1}{8} g_1^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{1}{8} g_1^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 - \frac{1}{6} a_t^3 \hat{\mu} - \frac{3}{16} \varepsilon \ln \frac{M_s^2}{\mu_M^2} a_t^4 + \frac{121}{576} g_2^2 a_t^2 \\
& + \frac{1}{4} \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}^2 - \frac{1}{4} \ln \frac{M_s^2}{\mu_M^2} a_t^4 + \frac{1}{4} \lambda_h \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{1}{4} \frac{g_2^2 a_t^2 \hat{\mu}_h^2}{\varepsilon} - \frac{1}{4} \frac{g_1^2 a_t^2}{\varepsilon} - \frac{1}{4} \frac{g_1^2 a_t^2 \hat{\mu}_h^2}{\varepsilon} \\
& - \frac{13}{48} a_t^2 \hat{\mu}^2 + \frac{11}{32} \varepsilon a_t^4 + \frac{5}{12} \kappa a_t^2 - \frac{1}{2} \kappa \ln \frac{M_s^2}{\mu_M^2} a_t \hat{\mu} - \frac{1}{2} \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 - \frac{1}{2} \frac{a_t^2 \hat{\mu}^2}{\varepsilon} + \frac{1}{2} \frac{a_t^4}{\varepsilon} \\
& \left. - \frac{1}{2} \frac{\lambda_h a_t^2}{\varepsilon} + \frac{13}{24} \kappa a_t \hat{\mu} - \frac{7}{8} \lambda_h a_t^2 + \frac{7}{6} a_t^4 + \frac{5}{4} \kappa a_t^2 \hat{\mu}_h^2 + \frac{\kappa a_t \hat{\mu}}{\varepsilon} + \frac{\kappa a_t^2 \hat{\mu}_h^2}{\varepsilon} \right) M_s^{-2} \\
& \varepsilon_{ik} \varepsilon_{jl} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} H_i H_j D_{\alpha_1}^{\dot{\alpha}_1} (H^\dagger)_k D_{\alpha_2}^{\dot{\alpha}_2} (H^\dagger)_l
\end{aligned} \tag{29}$$

term24

$$\begin{aligned}
& - \left(\frac{1}{96} dg_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{11}{576} dg_2^2 a_t^2 - \frac{1}{48} g_1^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{48} \frac{dg_2^2 a_t^2}{\varepsilon} \right. \\
& + \frac{1}{24} \frac{g_1^2 a_t^2}{\varepsilon} + \frac{17}{288} g_1^2 a_t^2 - \frac{19}{96} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{1}{4} g_2^2 a_t^2 \hat{\mu}_h^2 - \frac{1}{4} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 \\
& \left. + \frac{1}{4} \lambda_h a_t^2 + \frac{179}{576} g_2^2 a_t^2 + \frac{19}{48} \frac{g_2^2 a_t^2}{\varepsilon} + \frac{1}{2} \frac{g_2^2 a_t^2 \hat{\mu}_h^2}{\varepsilon} \right) M_s^{-2} \\
& \varepsilon_{il} \varepsilon_{jk} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} H_i H_j D_{\alpha_1}^{\dot{\alpha}_1} (H^\dagger)_k D_{\alpha_2}^{\dot{\alpha}_2} (H^\dagger)_l
\end{aligned} \tag{30}$$

nD: 0, H: 2, H+: 2

term26

$$\begin{aligned}
& + \left(a_t^3 \hat{\mu} - \ln \frac{M_s^2}{\mu_M^2} a_t^3 \hat{\mu} \hat{\mu}_h^2 - \frac{1}{4} a_t^2 \hat{\mu}^2 - \frac{1}{4} \kappa^2 \ln \frac{M_s^2}{\mu_M^2} + \frac{1}{4} \lambda_\varphi a_t^2 - \frac{1}{4} \lambda_\varphi \ln \frac{M_s^2}{\mu_M^2} a_t^2 \right. \\
& + \frac{1}{2} \kappa a_t \hat{\mu} + \frac{1}{2} \frac{\kappa^2}{\varepsilon} + \frac{1}{2} \frac{\lambda_\varphi a_t^2}{\varepsilon} + 2 a_t^3 \hat{\mu} \hat{\mu}_h^2 + 2 \ln \frac{M_s^2}{\mu_M^2} a_t^4 - 2 \kappa a_t^2 \hat{\mu}_h^4 + 2 \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^2 \\
& + 2 \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^4 + 2 \frac{a_t^3 \hat{\mu} \hat{\mu}_h^2}{\varepsilon} - 3 a_t^4 - 3 \kappa a_t^2 + 3 \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 + 3 \lambda_h a_t^2 \\
& - 3 \lambda_h \ln \frac{M_s^2}{\mu_M^2} a_t^2 - 4 \kappa a_t^2 \hat{\mu}_h^2 - 4 \frac{a_t^4}{\varepsilon} - 4 \frac{\kappa a_t^2}{\varepsilon} - 4 \frac{\kappa a_t^2 \hat{\mu}_h^4}{\varepsilon} + 6 \ln \frac{M_s^2}{\mu_M^2} a_t^4 \hat{\mu}_h^2 + 6 \lambda_h a_t^2 \hat{\mu}_h^2 \\
& \left. - 6 \lambda_h \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 - 6 \frac{\kappa a_t^2 \hat{\mu}_h^2}{\varepsilon} + 6 \frac{\lambda_h a_t^2}{\varepsilon} - 8 a_t^4 \hat{\mu}_h^2 - 12 \frac{a_t^4 \hat{\mu}_h^2}{\varepsilon} + 12 \frac{\lambda_h a_t^2 \hat{\mu}_h^2}{\varepsilon} \right) \\
& \varepsilon_{ik} \varepsilon_{jl} H_i H_j (H^\dagger)_k (H^\dagger)_l
\end{aligned} \tag{31}$$

nD: 2, WL: 1, H: 1, H+: 1

term22

$$\begin{aligned}
& + \left(\frac{1}{96} g_2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{48} \frac{g_2 a_t^2}{\varepsilon} - \frac{17}{576} g_2 a_t^2 \right) M_s^{-2} \\
& \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \varepsilon^{\alpha_1 \alpha_3} \varepsilon^{\alpha_2 \alpha_4} \delta_{ki} \delta_{lj} (W_L)_{\alpha_1 \alpha_2, ij} D_{\alpha_3}^{\dot{\alpha}_1} H_k D_{\alpha_4}^{\dot{\alpha}_2} (H^\dagger)_l
\end{aligned} \tag{32}$$

term22

$$+ \left(\frac{1}{96} g_2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{48} \frac{g_2 a_t^2}{\varepsilon} - \frac{17}{576} g_2 a_t^2 \right) M_s^{-2} \quad (33)$$

$$\varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \varepsilon^{\alpha_1 \alpha_4} \varepsilon^{\alpha_2 \alpha_3} \delta_{ki} \delta_{lj} (W_L)_{\alpha_1 \alpha_2, ij} D_{\alpha_3}^{\dot{\alpha}_1} H_k D_{\alpha_4}^{\dot{\alpha}_2} (H^\dagger)_l$$

term25

$$- \left(\frac{1}{96} g_2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{48} \frac{g_2 a_t^2}{\varepsilon} - \frac{17}{576} g_2 a_t^2 \right) M_s^{-2} \quad (34)$$

$$\varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} \varepsilon^{\alpha_1 \alpha_4} \varepsilon^{\alpha_2 \alpha_3} \delta_{ki} \delta_{lj} (W_L)_{\alpha_1 \alpha_2, ij} D_{\alpha_3}^{\dot{\alpha}_1} H_k D_{\alpha_4}^{\dot{\alpha}_2} (H^\dagger)_l$$

term25

$$- \left(\frac{1}{96} g_2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{48} \frac{g_2 a_t^2}{\varepsilon} - \frac{17}{576} g_2 a_t^2 \right) M_s^{-2} \quad (35)$$

$$\varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} \varepsilon^{\alpha_1 \alpha_3} \varepsilon^{\alpha_2 \alpha_4} \delta_{ki} \delta_{lj} (W_L)_{\alpha_1 \alpha_2, ij} D_{\alpha_3}^{\dot{\alpha}_1} H_k D_{\alpha_4}^{\dot{\alpha}_2} (H^\dagger)_l$$

nD: 2, H: 1, H+: 1, WL+: 1

term22

$$- \left(\frac{1}{96} g_2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{48} \frac{g_2 a_t^2}{\varepsilon} - \frac{17}{576} g_2 a_t^2 \right) M_s^{-2} \quad (36)$$

$$\varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_3 \dot{\alpha}_1} \varepsilon_{\dot{\alpha}_4 \dot{\alpha}_2} \delta_{ik} \delta_{jl} D_{\alpha_1}^{\dot{\alpha}_1} H_i D_{\alpha_2}^{\dot{\alpha}_2} (H^\dagger)_j (W_R)_{kl}^{\dot{\alpha}_3 \dot{\alpha}_4}$$

term22

$$- \left(\frac{1}{96} g_2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{48} \frac{g_2 a_t^2}{\varepsilon} - \frac{17}{576} g_2 a_t^2 \right) M_s^{-2} \quad (37)$$

$$\varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_3 \dot{\alpha}_2} \varepsilon_{\dot{\alpha}_4 \dot{\alpha}_1} \delta_{ik} \delta_{jl} D_{\alpha_1}^{\dot{\alpha}_1} H_i D_{\alpha_2}^{\dot{\alpha}_2} (H^\dagger)_j (W_R)_{kl}^{\dot{\alpha}_3 \dot{\alpha}_4}$$

term25

$$+ \left(\frac{1}{96} g_2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{48} \frac{g_2 a_t^2}{\varepsilon} - \frac{17}{576} g_2 a_t^2 \right) M_s^{-2} \quad (38)$$

$$\varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_3 \dot{\alpha}_2} \varepsilon_{\dot{\alpha}_4 \dot{\alpha}_1} \delta_{ik} \delta_{jl} D_{\alpha_1}^{\dot{\alpha}_1} H_i D_{\alpha_2}^{\dot{\alpha}_2} (H^\dagger)_j (W_R)_{kl}^{\dot{\alpha}_3 \dot{\alpha}_4}$$

term25

$$+ \left(\frac{1}{96} g_2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{48} \frac{g_2 a_t^2}{\varepsilon} - \frac{17}{576} g_2 a_t^2 \right) M_s^{-2} \quad (39)$$

$$\varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_3 \dot{\alpha}_1} \varepsilon_{\dot{\alpha}_4 \dot{\alpha}_2} \delta_{ik} \delta_{jl} D_{\alpha_1}^{\dot{\alpha}_1} H_i D_{\alpha_2}^{\dot{\alpha}_2} (H^\dagger)_j (W_R)_{kl}^{\dot{\alpha}_3 \dot{\alpha}_4}$$

nD: 0, H: 1, H+: 1, BL+: 2

term27

$$- \left(\frac{5}{576} g_1^2 a_t^2 - \frac{1}{96} g_1^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{1}{48} \frac{g_1^2 a_t^2}{\varepsilon} \right) M_s^{-2} \quad (40)$$

$$\varepsilon_{ij} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_3} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_4} H_i (H^\dagger)_j (B_R)^{\dot{\alpha}_1 \dot{\alpha}_2} (B_R)^{\dot{\alpha}_3 \dot{\alpha}_4}$$

term27

$$- \left(\frac{5}{576} g_1^2 a_t^2 - \frac{1}{96} g_1^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{1}{48} \frac{g_1^2 a_t^2}{\varepsilon} \right) M_s^{-2} \quad (41)$$

$$\varepsilon_{ij} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_4} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_3} H_i (H^\dagger)_j (B_R)^{\dot{\alpha}_1 \dot{\alpha}_2} (B_R)^{\dot{\alpha}_3 \dot{\alpha}_4}$$

nD: 0, BL: 2, H: 1, H+: 1

term27

$$-\left(\frac{5}{576}g_1^2a_t^2 - \frac{1}{96}g_1^2 \ln \frac{M_s^2}{\mu_M^2}a_t^2 + \frac{1}{48} \frac{g_1^2a_t^2}{\varepsilon}\right) M_s^{-2} \quad (42)$$

$$\varepsilon_{ij}\varepsilon^{\alpha_1\alpha_3}\varepsilon^{\alpha_2\alpha_4}(B_L)_{\alpha_1\alpha_2}(B_L)_{\alpha_3\alpha_4}H_i(H^\dagger)_j$$

term27

$$-\left(\frac{5}{576}g_1^2a_t^2 - \frac{1}{96}g_1^2 \ln \frac{M_s^2}{\mu_M^2}a_t^2 + \frac{1}{48} \frac{g_1^2a_t^2}{\varepsilon}\right) M_s^{-2} \quad (43)$$

$$\varepsilon_{ij}\varepsilon^{\alpha_1\alpha_4}\varepsilon^{\alpha_2\alpha_3}(B_L)_{\alpha_1\alpha_2}(B_L)_{\alpha_3\alpha_4}H_i(H^\dagger)_j$$

nD: 0, H: 1, H+: 1, BL+: 1, WL+: 1

term28

$$+\left(\frac{5}{288}g_1g_2a_t^2 - \frac{1}{48}g_1g_2 \ln \frac{M_s^2}{\mu_M^2}a_t^2 + \frac{1}{24} \frac{g_1g_2a_t^2}{\varepsilon}\right) M_s^{-2} \quad (44)$$

$$\varepsilon_{\dot{\alpha}_1\dot{\alpha}_3}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_4}\delta_{ik}\delta_{jl}H_i(H^\dagger)_j(B_R)^{\dot{\alpha}_1\dot{\alpha}_2}(W_R)_{kl}^{\dot{\alpha}_3\dot{\alpha}_4}$$

term28

$$+\left(\frac{5}{288}g_1g_2a_t^2 - \frac{1}{48}g_1g_2 \ln \frac{M_s^2}{\mu_M^2}a_t^2 + \frac{1}{24} \frac{g_1g_2a_t^2}{\varepsilon}\right) M_s^{-2} \quad (45)$$

$$\varepsilon_{\dot{\alpha}_1\dot{\alpha}_4}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_3}\delta_{ik}\delta_{jl}H_i(H^\dagger)_j(B_R)^{\dot{\alpha}_1\dot{\alpha}_2}(W_R)_{kl}^{\dot{\alpha}_3\dot{\alpha}_4}$$

nD: 0, BL: 1, WL: 1, H: 1, H+: 1

term28

$$+\left(\frac{5}{288}g_1g_2a_t^2 - \frac{1}{48}g_1g_2 \ln \frac{M_s^2}{\mu_M^2}a_t^2 + \frac{1}{24} \frac{g_1g_2a_t^2}{\varepsilon}\right) M_s^{-2} \quad (46)$$

$$\varepsilon^{\alpha_1\alpha_3}\varepsilon^{\alpha_2\alpha_4}\delta_{ki}\delta_{lj}(B_L)_{\alpha_1\alpha_2}(W_L)_{\alpha_3\alpha_4,ij}H_k(H^\dagger)_l$$

term28

$$+\left(\frac{5}{288}g_1g_2a_t^2 - \frac{1}{48}g_1g_2 \ln \frac{M_s^2}{\mu_M^2}a_t^2 + \frac{1}{24} \frac{g_1g_2a_t^2}{\varepsilon}\right) M_s^{-2} \quad (47)$$

$$\varepsilon^{\alpha_1\alpha_4}\varepsilon^{\alpha_2\alpha_3}\delta_{ki}\delta_{lj}(B_L)_{\alpha_1\alpha_2}(W_L)_{\alpha_3\alpha_4,ij}H_k(H^\dagger)_l$$

nD: 0, H: 3, H+: 3

term29

$$\begin{aligned}
& + \left(a_t^4 \hat{\mu}^2 - \kappa a_t^3 \hat{\mu} \hat{\mu}_h^4 + \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^3 \hat{\mu} + \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^3 \hat{\mu} \hat{\mu}_h^4 - \lambda_\varphi a_t^4 \hat{\mu}_h^2 \right. \\
& - \frac{1}{12} \kappa^3 - \frac{1}{12} \lambda_\varphi \ln \frac{M_s^2}{\mu_M^2} a_t^3 \hat{\mu} - \frac{1}{6} a_t^3 \hat{\mu}^3 + \frac{1}{6} \frac{\lambda_\varphi a_t^3 \hat{\mu}}{\varepsilon} - \frac{1}{4} \kappa^2 a_t \hat{\mu} \\
& + \frac{1}{4} \lambda_\varphi \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{1}{3} \lambda_\varphi a_t^3 \hat{\mu} + \frac{1}{2} \kappa a_t^2 \hat{\mu}^2 - \frac{1}{2} \lambda_\varphi a_t^4 + \frac{1}{2} \lambda_\varphi \ln \frac{M_s^2}{\mu_M^2} a_t^4 \hat{\mu}_h^2 \\
& - \frac{1}{2} \lambda_\varphi \kappa a_t^2 - \frac{1}{2} \frac{\lambda_\varphi \kappa a_t^2}{\varepsilon} - \frac{3}{2} \ln \frac{M_s^2}{\mu_M^2} a_t^4 \hat{\mu}^2 \hat{\mu}_h^2 + \frac{7}{2} a_t^4 \hat{\mu}^2 \hat{\mu}_h^2 + \frac{7}{2} \ln \frac{M_s^2}{\mu_M^2} a_t^5 \hat{\mu} \\
& + \frac{11}{2} \kappa^2 a_t^2 - \frac{15}{2} a_t^5 \hat{\mu} + \frac{19}{2} \kappa^2 a_t^2 \hat{\mu}_h^2 - \frac{21}{2} \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^4 + \frac{43}{3} a_t^6 + \frac{37}{2} \kappa a_t^4 \\
& - 2 \lambda_h \ln \frac{M_s^2}{\mu_M^2} a_t^3 \hat{\mu} - 2 \frac{\kappa a_t^3 \hat{\mu}}{\varepsilon} - 2 \frac{\kappa a_t^3 \hat{\mu} \hat{\mu}_h^4}{\varepsilon} + 3 a_t^6 \hat{\mu}_h^2 - 3 \ln \frac{M_s^2}{\mu_M^2} a_t^6 \hat{\mu}_h^2 \\
& + 3 \kappa^2 a_t^2 \hat{\mu}_h^4 - 3 \kappa^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 - 3 \kappa^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^4 + 3 \frac{a_t^4 \hat{\mu}^2 \hat{\mu}_h^2}{\varepsilon} \\
& + 4 \lambda_h a_t^3 \hat{\mu} + 4 \lambda_h^2 a_t^2 - 4 \lambda_h^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 + 4 \frac{\lambda_h a_t^3 \hat{\mu}}{\varepsilon} + 5 \ln \frac{M_s^2}{\mu_M^2} a_t^5 \hat{\mu} \hat{\mu}_h^2 \\
& - 5 \kappa a_t^3 \hat{\mu} + 6 \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^3 \hat{\mu} \hat{\mu}_h^2 - 6 \kappa^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 + 6 \frac{a_t^6 \hat{\mu}_h^2}{\varepsilon} + 6 \frac{\kappa^2 a_t^2}{\varepsilon} \\
& + 6 \frac{\kappa^2 a_t^2 \hat{\mu}_h^4}{\varepsilon} - 7 a_t^5 \hat{\mu} \hat{\mu}_h^2 - 7 \frac{a_t^5 \hat{\mu}}{\varepsilon} - 8 \lambda_h \ln \frac{M_s^2}{\mu_M^2} a_t^3 \hat{\mu} \hat{\mu}_h^2 + 8 \lambda_h \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^2 \\
& + 8 \frac{\lambda_h^2 a_t^2}{\varepsilon} - 9 \ln \frac{M_s^2}{\mu_M^2} a_t^6 - 10 \lambda_h \kappa a_t^2 - 10 \frac{a_t^5 \hat{\mu} \hat{\mu}_h^2}{\varepsilon} - 12 \kappa a_t^3 \hat{\mu} \hat{\mu}_h^2 \\
& + 12 \lambda_h a_t^3 \hat{\mu} \hat{\mu}_h^2 - 12 \lambda_h a_t^4 \hat{\mu}_h^2 + 12 \lambda_h \ln \frac{M_s^2}{\mu_M^2} a_t^4 + 12 \lambda_h \ln \frac{M_s^2}{\mu_M^2} a_t^4 \hat{\mu}_h^2 \\
& + 12 \lambda_h^2 a_t^2 \hat{\mu}_h^2 - 12 \lambda_h^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 - 12 \frac{\kappa a_t^3 \hat{\mu} \hat{\mu}_h^2}{\varepsilon} + 12 \frac{\kappa^2 a_t^2 \hat{\mu}_h^2}{\varepsilon} - 16 \lambda_h a_t^4 \\
& + 16 \frac{\lambda_h a_t^3 \hat{\mu} \hat{\mu}_h^2}{\varepsilon} - 16 \frac{\lambda_h \kappa a_t^2}{\varepsilon} + 18 \frac{a_t^6}{\varepsilon} + 20 \lambda_h \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 + 21 \frac{\kappa a_t^4}{\varepsilon} \\
& - 24 \lambda_h \kappa a_t^2 \hat{\mu}_h^2 - 24 \frac{\lambda_h a_t^4}{\varepsilon} - 24 \frac{\lambda_h a_t^4 \hat{\mu}_h^2}{\varepsilon} + 24 \frac{\lambda_h^2 a_t^2 \hat{\mu}_h^2}{\varepsilon} - 27 \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^4 \hat{\mu}_h^2 \\
& + 37 \kappa a_t^4 \hat{\mu}_h^2 - 40 \frac{\lambda_h \kappa a_t^2 \hat{\mu}_h^2}{\varepsilon} + 54 \frac{\kappa a_t^4 \hat{\mu}_h^2}{\varepsilon} - \frac{\lambda_\varphi a_t^4 \hat{\mu}_h^2}{\varepsilon} \Big) M_s^{-2} \\
& \varepsilon_{il} \varepsilon_{jm} \varepsilon_{kn} H_i H_j H_k (H^\dagger)_l (H^\dagger)_m (H^\dagger)_n
\end{aligned} \tag{48}$$

term30

$$\begin{aligned}
& - \left(\lambda_h \ln \frac{M_s^2}{\mu_M^2} a_t^3 \hat{\mu} - \lambda_h \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^2 - 2\lambda_h a_t^3 \hat{\mu} + 2\lambda_h \kappa a_t^2 - 2\lambda_h^2 a_t^2 \right. \\
& + 2\lambda_h^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 - 2\frac{\lambda_h a_t^3 \hat{\mu}}{\varepsilon} + 2\frac{\lambda_h \kappa a_t^2}{\varepsilon} + 4\lambda_h \ln \frac{M_s^2}{\mu_M^2} a_t^3 \hat{\mu} \hat{\mu}_h^2 \\
& - 4\lambda_h \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 - 4\frac{\lambda_h^2 a_t^2}{\varepsilon} - 6\lambda_h a_t^3 \hat{\mu} \hat{\mu}_h^2 + 6\lambda_h a_t^4 \hat{\mu}_h^2 - 6\lambda_h \ln \frac{M_s^2}{\mu_M^2} a_t^4 \\
& - 6\lambda_h \ln \frac{M_s^2}{\mu_M^2} a_t^4 \hat{\mu}_h^2 + 6\lambda_h \kappa a_t^2 \hat{\mu}_h^2 - 6\lambda_h^2 a_t^2 \hat{\mu}_h^2 + 6\lambda_h^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 + 8\lambda_h a_t^4 \\
& \left. - 8\frac{\lambda_h a_t^3 \hat{\mu} \hat{\mu}_h^2}{\varepsilon} + 8\frac{\lambda_h \kappa a_t^2 \hat{\mu}_h^2}{\varepsilon} + 12\frac{\lambda_h a_t^4}{\varepsilon} + 12\frac{\lambda_h a_t^4 \hat{\mu}_h^2}{\varepsilon} - 12\frac{\lambda_h^2 a_t^2 \hat{\mu}_h^2}{\varepsilon} \right) M_s^{-2} \\
& \varepsilon_{im} \varepsilon_{jl} \varepsilon_{kn} H_i H_j H_k (H^\dagger)_l (H^\dagger)_m (H^\dagger)_n
\end{aligned} \tag{49}$$

term31

$$\begin{aligned}
& + \left(2\lambda_h^2 a_t^2 - 2\lambda_h^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 + 4\frac{\lambda_h^2 a_t^2}{\varepsilon} + 6\lambda_h^2 a_t^2 \hat{\mu}_h^2 - 6\lambda_h^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 + 12\frac{\lambda_h^2 a_t^2 \hat{\mu}_h^2}{\varepsilon} \right) M_s^{-2} \\
& \varepsilon_{il} \varepsilon_{jn} \varepsilon_{km} H_i H_j H_k (H^\dagger)_l (H^\dagger)_m (H^\dagger)_n
\end{aligned} \tag{50}$$

term32

$$\begin{aligned}
& + \left(\frac{1}{2} \lambda_h^2 a_t^2 - \frac{1}{2} \lambda_h^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{3}{2} \lambda_h^2 a_t^2 \hat{\mu}_h^2 - \frac{3}{2} \lambda_h^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 + 3\frac{\lambda_h^2 a_t^2 \hat{\mu}_h^2}{\varepsilon} + \frac{\lambda_h^2 a_t^2}{\varepsilon} \right) M_s^{-2} \\
& \varepsilon_{in} \varepsilon_{jl} \varepsilon_{km} H_i H_j H_k (H^\dagger)_l (H^\dagger)_m (H^\dagger)_n
\end{aligned} \tag{51}$$

term33

$$\begin{aligned}
& + \left(\frac{1}{2} \lambda_h^2 a_t^2 - \frac{1}{2} \lambda_h^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{3}{2} \lambda_h^2 a_t^2 \hat{\mu}_h^2 - \frac{3}{2} \lambda_h^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 + 3\frac{\lambda_h^2 a_t^2 \hat{\mu}_h^2}{\varepsilon} + \frac{\lambda_h^2 a_t^2}{\varepsilon} \right) M_s^{-2} \\
& \varepsilon_{im} \varepsilon_{jn} \varepsilon_{kl} H_i H_j H_k (H^\dagger)_l (H^\dagger)_m (H^\dagger)_n
\end{aligned} \tag{52}$$

nD: 0, H: 1, H+: 1, WL+: 2

term34

$$\begin{aligned}
& + \left(\frac{5}{1152} g_2^2 a_t^2 - \frac{1}{192} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{1}{96} \frac{g_2^2 a_t^2}{\varepsilon} \right) M_s^{-2} \\
& \varepsilon_{ml} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_3} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_4} \delta_{ik} \delta_{jn} H_i (H^\dagger)_j (W_R)_{kl}^{\dot{\alpha}_1 \dot{\alpha}_2} (W_R)_{mn}^{\dot{\alpha}_3 \dot{\alpha}_4}
\end{aligned} \tag{53}$$

term34

$$\begin{aligned}
& + \left(\frac{5}{1152} g_2^2 a_t^2 - \frac{1}{192} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{1}{96} \frac{g_2^2 a_t^2}{\varepsilon} \right) M_s^{-2} \\
& \varepsilon_{ml} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_4} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_3} \delta_{ik} \delta_{jn} H_i (H^\dagger)_j (W_R)_{kl}^{\dot{\alpha}_1 \dot{\alpha}_2} (W_R)_{mn}^{\dot{\alpha}_3 \dot{\alpha}_4}
\end{aligned} \tag{54}$$

term35

$$\begin{aligned}
& - \left(\frac{5}{1152} g_2^2 a_t^2 - \frac{1}{192} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{1}{96} \frac{g_2^2 a_t^2}{\varepsilon} \right) M_s^{-2} \\
& \varepsilon_{nk} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_3} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_4} \delta_{im} \delta_{jl} H_i (H^\dagger)_j (W_R)_{kl}^{\dot{\alpha}_1 \dot{\alpha}_2} (W_R)_{mn}^{\dot{\alpha}_3 \dot{\alpha}_4}
\end{aligned} \tag{55}$$

term35

$$\begin{aligned}
& - \left(\frac{5}{1152} g_2^2 a_t^2 - \frac{1}{192} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{1}{96} \frac{g_2^2 a_t^2}{\varepsilon} \right) M_s^{-2} \\
& \varepsilon_{nk} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_4} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_3} \delta_{im} \delta_{jl} H_i (H^\dagger)_j (W_R)_{kl}^{\dot{\alpha}_1 \dot{\alpha}_2} (W_R)_{mn}^{\dot{\alpha}_3 \dot{\alpha}_4}
\end{aligned} \tag{56}$$

nD: 0, WL: 2, H: 1, H+: 1

term34

$$\begin{aligned}
& + \left(\frac{5}{1152} g_2^2 a_t^2 - \frac{1}{192} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{1}{96} \frac{g_2^2 a_t^2}{\varepsilon} \right) M_s^{-2} \\
& \varepsilon_{kj} \varepsilon^{\alpha_1 \alpha_3} \varepsilon^{\alpha_2 \alpha_4} \delta_{mi} \delta_{nl} (W_L)_{\alpha_1 \alpha_2, ij} (W_L)_{\alpha_3 \alpha_4, kl} H_m (H^\dagger)_n
\end{aligned} \tag{57}$$

term34

$$\begin{aligned}
& + \left(\frac{5}{1152} g_2^2 a_t^2 - \frac{1}{192} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{1}{96} \frac{g_2^2 a_t^2}{\varepsilon} \right) M_s^{-2} \\
& \varepsilon_{kj} \varepsilon^{\alpha_1 \alpha_4} \varepsilon^{\alpha_2 \alpha_3} \delta_{mi} \delta_{nl} (W_L)_{\alpha_1 \alpha_2, ij} (W_L)_{\alpha_3 \alpha_4, kl} H_m (H^\dagger)_n
\end{aligned} \tag{58}$$

term35

$$\begin{aligned}
& - \left(\frac{5}{1152} g_2^2 a_t^2 - \frac{1}{192} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{1}{96} \frac{g_2^2 a_t^2}{\varepsilon} \right) M_s^{-2} \\
& \varepsilon_{li} \varepsilon^{\alpha_1 \alpha_3} \varepsilon^{\alpha_2 \alpha_4} \delta_{mk} \delta_{nj} (W_L)_{\alpha_1 \alpha_2, ij} (W_L)_{\alpha_3 \alpha_4, kl} H_m (H^\dagger)_n
\end{aligned} \tag{59}$$

term35

$$\begin{aligned}
& - \left(\frac{5}{1152} g_2^2 a_t^2 - \frac{1}{192} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{1}{96} \frac{g_2^2 a_t^2}{\varepsilon} \right) M_s^{-2} \\
& \varepsilon_{li} \varepsilon^{\alpha_1 \alpha_4} \varepsilon^{\alpha_2 \alpha_3} \delta_{mk} \delta_{nj} (W_L)_{\alpha_1 \alpha_2, ij} (W_L)_{\alpha_3 \alpha_4, kl} H_m (H^\dagger)_n
\end{aligned} \tag{60}$$

nD: 1, L: 1, H: 1, H+: 1, L+: 1

term38

$$\begin{aligned}
& - \left(\frac{1}{4} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{2} \frac{a_t^2}{\varepsilon} - \frac{5}{8} a_t^2 \right) M_s^{-2} \\
& (y_e)^{wu} (y_e^\dagger)^{uv} \varepsilon_{ik} \varepsilon_{jl} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} L_{\alpha_1, i}^v D_{\alpha_2}^{\dot{\alpha}_1} H_j (H^\dagger)_k (L^\dagger)_l^{\dot{\alpha}_2, w}
\end{aligned} \tag{61}$$

term43

$$\begin{aligned}
& + \left(\frac{1}{4} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{2} \frac{a_t^2}{\varepsilon} - \frac{5}{8} a_t^2 \right) M_s^{-2} \\
& (y_e)^{wu} (y_e^\dagger)^{uv} \varepsilon_{ik} \varepsilon_{jl} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} L_{\alpha_1, i}^v H_j D_{\alpha_2}^{\dot{\alpha}_1} (H^\dagger)_k (L^\dagger)_l^{\dot{\alpha}_2, w}
\end{aligned} \tag{62}$$

term48

$$\begin{aligned}
& - \left(\frac{1}{4} a_t^2 \right) M_s^{-2} \\
& (y_e)^{wu} (y_e^\dagger)^{uv} \varepsilon_{ik} \varepsilon_{jl} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} D_{\alpha_1}^{\dot{\alpha}_1} L_{\alpha_2, i}^v H_j (H^\dagger)_k (L^\dagger)_l^{\dot{\alpha}_2, w}
\end{aligned} \tag{63}$$

term53

$$\begin{aligned}
& - \left(\frac{1}{4} a_t^2 \right) M_s^{-2} \\
& (y_e)^{wu} (y_e^\dagger)^{uv} \varepsilon_{ik} \varepsilon_{jl} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} L_{\alpha_1, i}^v H_j (H^\dagger)_k D_{\alpha_2}^{\dot{\alpha}_1} (L^\dagger)_l^{\dot{\alpha}_2, w}
\end{aligned} \tag{64}$$

term58

$$\begin{aligned}
& - \left(\frac{1}{4} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{8} a_t^2 - \frac{1}{2} \frac{a_t^2}{\varepsilon} \right) M_s^{-2} \\
& (y_e)^{wu} (y_e^\dagger)^{uv} \varepsilon_{ik} \varepsilon_{jl} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} L_{\alpha_1, i}^v H_j (H^\dagger)_k D_{\dot{\alpha}_2}^{\dot{\alpha}_1} (L^\dagger)_l^{\dot{\alpha}_2, w}
\end{aligned} \tag{65}$$

term63

$$\begin{aligned}
& - \left(\frac{1}{4} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{8} a_t^2 - \frac{1}{2} \frac{a_t^2}{\varepsilon} \right) M_s^{-2} \\
& (y_e)^{wu} (y_e^\dagger)^{uv} \varepsilon_{ik} \varepsilon_{jl} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} D_{\alpha_1}^{\dot{\alpha}_1} L_{\alpha_2, i}^v H_j (H^\dagger)_k (L^\dagger)_l^{\dot{\alpha}_2, w}
\end{aligned} \tag{66}$$

nD: 1, Q: 1, H: 1, H+: 1, Q+: 1

term39

$$\begin{aligned}
& - \left(\frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{4} \frac{a_t^2}{\varepsilon} - \frac{5}{16} a_t^2 \right) M_s^{-2} \\
& (y_d)^{wu} (y_d^\dagger)^{uv} \varepsilon_{ik} \varepsilon_{jl} \varepsilon_{abc} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} Q_{\alpha_1, i, a}^v D_{\alpha_2}^{\dot{\alpha}_1} H_j (H^\dagger)_k (Q^\dagger)_{l, bc}^{\dot{\alpha}_2, w}
\end{aligned} \tag{67}$$

term44

$$\begin{aligned}
& + \left(\frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{4} \frac{a_t^2}{\varepsilon} - \frac{5}{16} a_t^2 \right) M_s^{-2} \\
& (y_d)^{wu} (y_d^\dagger)^{uv} \varepsilon_{ik} \varepsilon_{jl} \varepsilon_{abc} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} Q_{\alpha_1, i, a}^v H_j D_{\alpha_2}^{\dot{\alpha}_1} (H^\dagger)_k (Q^\dagger)_{l, bc}^{\dot{\alpha}_2, w}
\end{aligned} \tag{68}$$

term49

$$\begin{aligned}
& - \left(\frac{1}{8} a_t^2 \right) M_s^{-2} \\
& (y_d)^{wu} (y_d^\dagger)^{uv} \varepsilon_{ik} \varepsilon_{jl} \varepsilon_{abc} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} D_{\alpha_1}^{\dot{\alpha}_1} Q_{\alpha_2, i, a}^v H_j (H^\dagger)_k (Q^\dagger)_{l, bc}^{\dot{\alpha}_2, w}
\end{aligned} \tag{69}$$

term54

$$\begin{aligned}
& - \left(\frac{1}{8} a_t^2 \right) M_s^{-2} \\
& (y_d)^{wu} (y_d^\dagger)^{uv} \varepsilon_{ik} \varepsilon_{jl} \varepsilon_{abc} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} Q_{\alpha_1, i, a}^v H_j (H^\dagger)_k D_{\alpha_2}^{\dot{\alpha}_1} (Q^\dagger)_{l, bc}^{\dot{\alpha}_2, w}
\end{aligned} \tag{70}$$

term59

$$\begin{aligned}
& - \left(\frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{16} a_t^2 - \frac{1}{4} \frac{a_t^2}{\varepsilon} \right) M_s^{-2} \\
& (y_d)^{wu} (y_d^\dagger)^{uv} \varepsilon_{ik} \varepsilon_{jl} \varepsilon_{abc} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} Q_{\alpha_1, i, a}^v H_j (H^\dagger)_k D_{\alpha_2}^{\dot{\alpha}_1} (Q^\dagger)_{l, bc}^{\dot{\alpha}_2, w}
\end{aligned} \tag{71}$$

term64

$$\begin{aligned}
& - \left(\frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{16} a_t^2 - \frac{1}{4} \frac{a_t^2}{\varepsilon} \right) M_s^{-2} \\
& (y_d)^{wu} (y_d^\dagger)^{uv} \varepsilon_{ik} \varepsilon_{jl} \varepsilon_{abc} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} D_{\alpha_1}^{\dot{\alpha}_1} Q_{\alpha_2, i, a}^v H_j (H^\dagger)_k (Q^\dagger)_{l, bc}^{\dot{\alpha}_2, w}
\end{aligned} \tag{72}$$

term88

$$\begin{aligned}
& + \left(\frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{4} \frac{a_t^2}{\varepsilon} - \frac{5}{16} a_t^2 \right) M_s^{-2} \\
& (y_u)^{wu} (y_u^\dagger)^{uv} \varepsilon_{ij} \varepsilon_{kl} \varepsilon_{abc} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} Q_{\alpha_1, i, a}^v D_{\alpha_2}^{\dot{\alpha}_1} H_j (H^\dagger)_k (Q^\dagger)_{l, bc}^{\dot{\alpha}_2, w}
\end{aligned} \tag{73}$$

term89

$$\begin{aligned}
& - \left(\frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{4} \frac{a_t^2}{\varepsilon} - \frac{5}{16} a_t^2 \right) M_s^{-2} \\
& (y_u)^{wu} (y_u^\dagger)^{uv} \varepsilon_{ji} \varepsilon_{lk} \varepsilon_{abc} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} Q_{\alpha_1, i, a}^v H_j D_{\alpha_2}^{\dot{\alpha}_1} (H^\dagger)_k (Q^\dagger)_{l, bc}^{\dot{\alpha}_2, w}
\end{aligned} \tag{74}$$

term90

$$\begin{aligned}
& - \left(\frac{1}{8} a_t^2 \right) M_s^{-2} \\
& (y_u)^{wu} (y_u^\dagger)^{uv} \varepsilon_{ji} \varepsilon_{kl} \varepsilon_{abc} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} D_{\alpha_1}^{\dot{\alpha}_1} Q_{\alpha_2, i, a}^v H_j (H^\dagger)_k (Q^\dagger)_{l, bc}^{\dot{\alpha}_2, w}
\end{aligned} \tag{75}$$

term91

$$\begin{aligned}
& - \left(\frac{1}{8} a_t^2 \right) M_s^{-2} \\
& (y_u)^{wu} (y_u^\dagger)^{uv} \varepsilon_{ji} \varepsilon_{lk} \varepsilon_{abc} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} Q_{\alpha_1, i, a}^v H_j (H^\dagger)_k D_{\alpha_2}^{\dot{\alpha}_1} (Q^\dagger)_{l, bc}^{\dot{\alpha}_2, w}
\end{aligned} \tag{76}$$

term92

$$\begin{aligned}
& + \left(\frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{16} a_t^2 - \frac{1}{4} \frac{a_t^2}{\varepsilon} \right) M_s^{-2} \\
& (y_u)^{wu} (y_u^\dagger)^{uv} \varepsilon_{ij} \varepsilon_{lk} \varepsilon_{abc} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} Q_{\alpha_1, i, a}^v H_j (H^\dagger)_k D_{\alpha_2}^{\dot{\alpha}_1} (Q^\dagger)_{l, bc}^{\dot{\alpha}_2, w}
\end{aligned} \tag{77}$$

term93

$$\begin{aligned}
& + \left(\frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{16} a_t^2 - \frac{1}{4} \frac{a_t^2}{\varepsilon} \right) M_s^{-2} \\
& (y_u)^{wu} (y_u^\dagger)^{uv} \varepsilon_{ji} \varepsilon_{kl} \varepsilon_{abc} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} D_{\alpha_1}^{\dot{\alpha}_1} Q_{\alpha_2, i, a}^v H_j (H^\dagger)_k (Q^\dagger)_{l, bc}^{\dot{\alpha}_2, w}
\end{aligned} \tag{78}$$

nD: 1, eC: 1, H: 1, H+: 1, eC+: 1

term40

$$\begin{aligned}
& - \left(\frac{1}{4} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{2} \frac{a_t^2}{\varepsilon} - \frac{5}{8} a_t^2 \right) M_s^{-2} \\
& (y_e)^{uw} (y_e^\dagger)^{vu} \varepsilon_{ij} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} (e_C)_{\alpha_1}^v D_{\alpha_2}^{\dot{\alpha}_1} H_i (H^\dagger)_j (e_C^\dagger)^{\dot{\alpha}_2, w}
\end{aligned} \tag{79}$$

term46

$$\begin{aligned}
& + \left(\frac{1}{4} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{2} \frac{a_t^2}{\varepsilon} - \frac{5}{8} a_t^2 \right) M_s^{-2} \\
& (y_e)^{uw} (y_e^\dagger)^{vu} \varepsilon_{ij} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} (e_C)_{\alpha_1}^v H_i D_{\alpha_2}^{\dot{\alpha}_1} (H^\dagger)_j (e_C^\dagger)^{\dot{\alpha}_2, w}
\end{aligned} \tag{80}$$

term51

$$\begin{aligned}
& - \left(\frac{1}{4} a_t^2 \right) M_s^{-2} \\
& (y_e)^{uw} (y_e^\dagger)^{vu} \varepsilon_{ij} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} (e_C)_{\alpha_1}^v H_i (H^\dagger)_j D_{\alpha_2}^{\dot{\alpha}_1} (e_C^\dagger)^{\dot{\alpha}_2, w}
\end{aligned} \tag{81}$$

term55

$$\begin{aligned}
& + \left(\frac{1}{4} a_t^2 \right) M_s^{-2} \\
& (y_e)^{uw} (y_e^\dagger)^{vu} \varepsilon_{ij} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} D_{\alpha_1}^{\dot{\alpha}_1} (e_C)_{\alpha_2}^v H_i (H^\dagger)_j (e_C^\dagger)^{\dot{\alpha}_2, w}
\end{aligned} \tag{82}$$

term61

$$+ \left(\frac{1}{4} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{8} a_t^2 - \frac{1}{2} \frac{a_t^2}{\varepsilon} \right) M_s^{-2} \quad (83)$$

$$(y_e)^{uw} (y_e^\dagger)^{vu} \varepsilon_{ij} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} D_{\alpha_1}^{\dot{\alpha}_1} (e_{\mathbb{C}})^v_{\alpha_2} H_i (H^\dagger)_j (e_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}_{\dot{\alpha}_1}$$

term65

$$- \left(\frac{1}{4} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{8} a_t^2 - \frac{1}{2} \frac{a_t^2}{\varepsilon} \right) M_s^{-2} \quad (84)$$

$$(y_e)^{uw} (y_e^\dagger)^{vu} \varepsilon_{ij} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} (e_{\mathbb{C}})^v_{\alpha_1} H_i (H^\dagger)_j D_{\alpha_2}^{\dot{\alpha}_1} (e_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}_{\dot{\alpha}_2}$$

nD: 1, dC: 1, H: 1, H+: 1, dC+: 1

term41

$$- \left(\frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{4} \frac{a_t^2}{\varepsilon} - \frac{5}{16} a_t^2 \right) M_s^{-2} \quad (85)$$

$$(y_d)^{uw} (y_d^\dagger)^{vu} \varepsilon_{ij} \varepsilon_{cab} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} (d_{\mathbb{C}})^v_{\alpha_1, ab} D_{\alpha_2}^{\dot{\alpha}_1} H_i (H^\dagger)_j (d_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}_c$$

term47

$$+ \left(\frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{4} \frac{a_t^2}{\varepsilon} - \frac{5}{16} a_t^2 \right) M_s^{-2} \quad (86)$$

$$(y_d)^{uw} (y_d^\dagger)^{vu} \varepsilon_{ij} \varepsilon_{cab} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} (d_{\mathbb{C}})^v_{\alpha_1, ab} H_i D_{\alpha_2}^{\dot{\alpha}_1} (H^\dagger)_j (d_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}_c$$

term52

$$- \left(\frac{1}{8} a_t^2 \right) M_s^{-2} \quad (87)$$

$$(y_d)^{uw} (y_d^\dagger)^{vu} \varepsilon_{ij} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} (d_{\mathbb{C}})^v_{\alpha_1, ab} H_i (H^\dagger)_j D_{\alpha_2}^{\dot{\alpha}_1} (d_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}_c$$

term56

$$+ \left(\frac{1}{8} a_t^2 \right) M_s^{-2} \quad (88)$$

$$(y_d)^{uw} (y_d^\dagger)^{vu} \varepsilon_{ij} \varepsilon_{cab} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} D_{\alpha_1}^{\dot{\alpha}_1} (d_{\mathbb{C}})^v_{\alpha_2, ab} H_i (H^\dagger)_j (d_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}_c$$

term62

$$+ \left(\frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{16} a_t^2 - \frac{1}{4} \frac{a_t^2}{\varepsilon} \right) M_s^{-2} \quad (89)$$

$$(y_d)^{uw} (y_d^\dagger)^{vu} \varepsilon_{ij} \varepsilon_{cab} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} D_{\alpha_1}^{\dot{\alpha}_1} (d_{\mathbb{C}})^v_{\alpha_2, ab} H_i (H^\dagger)_j (d_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}_c$$

term66

$$- \left(\frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{16} a_t^2 - \frac{1}{4} \frac{a_t^2}{\varepsilon} \right) M_s^{-2} \quad (90)$$

$$(y_d)^{uw} (y_d^\dagger)^{vu} \varepsilon_{ij} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} (d_{\mathbb{C}})^v_{\alpha_1, ab} H_i (H^\dagger)_j D_{\alpha_2}^{\dot{\alpha}_1} (d_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}_c$$

nD: 1, uC: 1, H: 1, H+: 1, uC+: 1

term42

$$+ \left(\frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{4} \frac{a_t^2}{\varepsilon} - \frac{5}{16} a_t^2 \right) M_s^{-2} \quad (91)$$

$$(y_u)^{uw} (y_u^\dagger)^{vu} \varepsilon_{ij} \varepsilon_{cab} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} (u_{\mathbb{C}})^v_{\alpha_1, ab} D_{\alpha_2}^{\dot{\alpha}_1} H_i (H^\dagger)_j (u_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}_c$$

term45

$$\begin{aligned}
& - \left(\frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{4} \frac{a_t^2}{\varepsilon} - \frac{5}{16} a_t^2 \right) M_s^{-2} \\
& (y_u)^{uw} (y_u^\dagger)^{vu} \varepsilon_{ij} \varepsilon_{cab} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} (u_{\mathbb{C}})^v_{\alpha_1, ab} H_i D_{\alpha_2}^{\dot{\alpha}_1} (H^\dagger)_j (u_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}_c
\end{aligned} \tag{92}$$

term50

$$\begin{aligned}
& + \left(\frac{1}{8} a_t^2 \right) M_s^{-2} \\
& (y_u)^{uw} (y_u^\dagger)^{vu} \varepsilon_{ij} \varepsilon_{cab} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} (u_{\mathbb{C}})^v_{\alpha_1, ab} H_i (H^\dagger)_j D_{\alpha_2}^{\dot{\alpha}_1} (u_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}_c
\end{aligned} \tag{93}$$

term57

$$\begin{aligned}
& + \left(\frac{1}{8} a_t^2 \right) M_s^{-2} \\
& (y_u)^{uw} (y_u^\dagger)^{vu} \varepsilon_{ij} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} D_{\alpha_1}^{\dot{\alpha}_1} (u_{\mathbb{C}})^v_{\alpha_2, ab} H_i (H^\dagger)_j (u_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}_c
\end{aligned} \tag{94}$$

term60

$$\begin{aligned}
& + \left(\frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{16} a_t^2 - \frac{1}{4} \frac{a_t^2}{\varepsilon} \right) M_s^{-2} \\
& (y_u)^{uw} (y_u^\dagger)^{vu} \varepsilon_{ij} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} D_{\alpha_1}^{\dot{\alpha}_1} (u_{\mathbb{C}})^v_{\alpha_2, ab} H_i (H^\dagger)_j (u_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}_c
\end{aligned} \tag{95}$$

term67

$$\begin{aligned}
& + \left(\frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{16} a_t^2 - \frac{1}{4} \frac{a_t^2}{\varepsilon} \right) M_s^{-2} \\
& (y_u)^{uw} (y_u^\dagger)^{vu} \varepsilon_{ij} \varepsilon_{cab} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} (u_{\mathbb{C}})^v_{\alpha_1, ab} H_i (H^\dagger)_j D_{\alpha_2}^{\dot{\alpha}_1} (u_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}_c
\end{aligned} \tag{96}$$

nD: 1, uC: 1, H: 2, dC+: 1

term68

$$\begin{aligned}
& + \left(\frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{4} \frac{a_t^2}{\varepsilon} - \frac{5}{16} a_t^2 \right) M_s^{-2} \\
& (y_u^\dagger)^{vu} (y_d)^{uw} \varepsilon_{ji} \varepsilon_{cab} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} (u_{\mathbb{C}})^v_{\alpha_1, ab} D_{\alpha_2}^{\dot{\alpha}_1} H_i H_j (d_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}_c
\end{aligned} \tag{97}$$

term69

$$\begin{aligned}
& + \left(\frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{4} \frac{a_t^2}{\varepsilon} - \frac{5}{16} a_t^2 \right) M_s^{-2} \\
& (y_u^\dagger)^{vu} (y_d)^{uw} \varepsilon_{ji} \varepsilon_{cab} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} (u_{\mathbb{C}})^v_{\alpha_1, ab} D_{\alpha_2}^{\dot{\alpha}_1} H_i H_j (d_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}_c
\end{aligned} \tag{98}$$

term73

$$\begin{aligned}
& - \left(\frac{1}{8} a_t^2 \right) M_s^{-2} \\
& (y_u^\dagger)^{vu} (y_d)^{uw} \varepsilon_{ji} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} (u_{\mathbb{C}})^v_{\alpha_1, ab} H_i H_j D_{\alpha_2}^{\dot{\alpha}_1} (d_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}_c
\end{aligned} \tag{99}$$

term74

$$\begin{aligned}
& + \left(\frac{1}{8} a_t^2 \right) M_s^{-2} \\
& (y_u^\dagger)^{vu} (y_d)^{uw} \varepsilon_{ji} \varepsilon_{cab} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} D_{\alpha_1}^{\dot{\alpha}_1} (u_{\mathbb{C}})^v_{\alpha_2, ab} H_i H_j (d_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}_c
\end{aligned} \tag{100}$$

term77

$$\begin{aligned}
& - \left(\frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{16} a_t^2 - \frac{1}{4} \frac{a_t^2}{\varepsilon} \right) M_s^{-2} \\
& (y_u^\dagger)^{vu} (y_d)^{uw} \varepsilon_{ji} \varepsilon_{cab} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} D_{\alpha_1}^{\dot{\alpha}_1} (u_{\mathbb{C}})^v_{\alpha_2, ab} H_i H_j (d_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}_c
\end{aligned} \tag{101}$$

term78

$$\begin{aligned}
& - \left(\frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{16} a_t^2 - \frac{1}{4} \frac{a_t^2}{\varepsilon} \right) M_s^{-2} \\
& (y_u^\dagger)^{vu} (y_d)^{uw} \varepsilon_{ji} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} (u_{\mathbb{C}})^v_{\alpha_1, ab} H_i H_j D_{\alpha_2}^{\dot{\alpha}_1} (d_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}_c
\end{aligned} \tag{102}$$

nD: 1, dC: 1, H+: 2, uC+: 1

term70

$$\begin{aligned}
& - \left(\frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{4} \frac{a_t^2}{\varepsilon} - \frac{5}{16} a_t^2 \right) M_s^{-2} \\
& (y_u)^{uw} (y_d^\dagger)^{vu} \varepsilon_{ij} \varepsilon_{cab} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} (d_{\mathbb{C}})^v_{\alpha_1, ab} D_{\alpha_2}^{\dot{\alpha}_1} (H^\dagger)_i (H^\dagger)_j (u_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}_c
\end{aligned} \tag{103}$$

term71

$$\begin{aligned}
& + \left(\frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{4} \frac{a_t^2}{\varepsilon} - \frac{5}{16} a_t^2 \right) M_s^{-2} \\
& (y_u)^{uw} (y_d^\dagger)^{vu} \varepsilon_{ji} \varepsilon_{cab} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} (d_{\mathbb{C}})^v_{\alpha_1, ab} D_{\alpha_2}^{\dot{\alpha}_1} (H^\dagger)_i (H^\dagger)_j (u_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}_c
\end{aligned} \tag{104}$$

term72

$$\begin{aligned}
& - \left(\frac{1}{8} a_t^2 \right) M_s^{-2} \\
& (y_u)^{uw} (y_d^\dagger)^{vu} \varepsilon_{ji} \varepsilon_{cab} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} (d_{\mathbb{C}})^v_{\alpha_1, ab} (H^\dagger)_i (H^\dagger)_j D_{\alpha_2}^{\dot{\alpha}_1} (u_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}_c
\end{aligned} \tag{105}$$

term75

$$\begin{aligned}
& + \left(\frac{1}{8} a_t^2 \right) M_s^{-2} \\
& (y_u)^{uw} (y_d^\dagger)^{vu} \varepsilon_{ji} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} D_{\alpha_1}^{\dot{\alpha}_1} (d_{\mathbb{C}})^v_{\alpha_2, ab} (H^\dagger)_i (H^\dagger)_j (u_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}_c
\end{aligned} \tag{106}$$

term76

$$\begin{aligned}
& + \left(\frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{16} a_t^2 - \frac{1}{4} \frac{a_t^2}{\varepsilon} \right) M_s^{-2} \\
& (y_u)^{uw} (y_d^\dagger)^{vu} \varepsilon_{ij} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} D_{\alpha_1}^{\dot{\alpha}_1} (d_{\mathbb{C}})^v_{\alpha_2, ab} (H^\dagger)_i (H^\dagger)_j (u_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}_c
\end{aligned} \tag{107}$$

term79

$$\begin{aligned}
& + \left(\frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{16} a_t^2 - \frac{1}{4} \frac{a_t^2}{\varepsilon} \right) M_s^{-2} \\
& (y_u)^{uw} (y_d^\dagger)^{vu} \varepsilon_{ji} \varepsilon_{cab} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} (d_{\mathbb{C}})^v_{\alpha_1, ab} (H^\dagger)_i (H^\dagger)_j D_{\alpha_2}^{\dot{\alpha}_1} (u_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}_c
\end{aligned} \tag{108}$$

nD: 0, L: 1, eC: 1, H: 1, H+: 2

term80

$$\begin{aligned}
& + \left(\frac{1}{8} d \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{16} d a_t^2 - \frac{1}{4} \frac{d a_t^2}{\varepsilon} \right) M_s^{-2} \\
& (y_e)^{uv} (y_e^\dagger)^{xu} (y_e^\dagger)^{vw} \varepsilon_{il} \varepsilon_{jk} \varepsilon^{\alpha_1 \alpha_2} L_{\alpha_1, i}^w (e_{\mathbb{C}})^x_{\alpha_2} H_j (H^\dagger)_k (H^\dagger)_l
\end{aligned} \tag{109}$$

term86

$$+ \left(\frac{1}{8} d \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{16} da_t^2 - \frac{1}{4} \frac{da_t^2}{\varepsilon} \right) M_s^{-2} \quad (110)$$

$$(y_e)^{uv} (y_e^\dagger)^{xu} (y_e^\dagger)^{vw} \varepsilon_{jk} \varepsilon_{il} \varepsilon^{\alpha_1 \alpha_2} L_{\alpha_1, i}^w (e_{\mathbb{C}})_{\alpha_2}^x H_j (H^\dagger)_k (H^\dagger)_l$$

nD: 0, Q: 1, dC: 1, H: 1, H+: 2

term81

$$+ \left(\frac{1}{16} d \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{32} da_t^2 - \frac{1}{8} \frac{da_t^2}{\varepsilon} \right) M_s^{-2} \quad (111)$$

$$(y_d)^{uv} (y_d^\dagger)^{xu} (y_d^\dagger)^{vw} \varepsilon_{il} \varepsilon_{jk} \varepsilon_{abc} \varepsilon^{\alpha_1 \alpha_2} Q_{\alpha_1, i, a}^w (d_{\mathbb{C}})_{\alpha_2, bc}^x H_j (H^\dagger)_k (H^\dagger)_l$$

term87

$$+ \left(\frac{1}{16} d \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{32} da_t^2 - \frac{1}{8} \frac{da_t^2}{\varepsilon} \right) M_s^{-2} \quad (112)$$

$$(y_d)^{uv} (y_d^\dagger)^{xu} (y_d^\dagger)^{vw} \varepsilon_{jk} \varepsilon_{il} \varepsilon_{abc} \varepsilon^{\alpha_1 \alpha_2} Q_{\alpha_1, i, a}^w (d_{\mathbb{C}})_{\alpha_2, bc}^x H_j (H^\dagger)_k (H^\dagger)_l$$

term102

$$- \left(\frac{1}{16} d \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{32} da_t^2 - \frac{1}{8} \frac{da_t^2}{\varepsilon} \right) M_s^{-2} \quad (113)$$

$$(y_u)^{vu} (y_u^\dagger)^{uw} (y_d^\dagger)^{xv} \varepsilon_{ji} \varepsilon_{kl} \varepsilon_{abc} \varepsilon^{\alpha_1 \alpha_2} Q_{\alpha_1, i, a}^w (d_{\mathbb{C}})_{\alpha_2, bc}^x H_j (H^\dagger)_k (H^\dagger)_l$$

term105

$$- \left(\frac{1}{16} d \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{32} da_t^2 - \frac{1}{8} \frac{da_t^2}{\varepsilon} \right) M_s^{-2} \quad (114)$$

$$(y_u)^{vu} (y_u^\dagger)^{uw} (y_d^\dagger)^{xv} \varepsilon_{ij} \varepsilon_{kl} \varepsilon_{abc} \varepsilon^{\alpha_1 \alpha_2} Q_{\alpha_1, i, a}^w (d_{\mathbb{C}})_{\alpha_2, bc}^x H_j (H^\dagger)_k (H^\dagger)_l$$

nD: 0, H: 2, H+: 1, L+: 1, eC+: 1

term82

$$- \left(\frac{1}{8} d \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{16} da_t^2 - \frac{1}{4} \frac{da_t^2}{\varepsilon} \right) M_s^{-2} \quad (115)$$

$$(y_e)^{wu} (y_e)^{vx} (y_e^\dagger)^{uv} \varepsilon_{il} \varepsilon_{jk} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} H_i H_j (H^\dagger)_k (L^\dagger)_l^{\dot{\alpha}_1, w} (e_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, x}$$

term84

$$- \left(\frac{1}{8} d \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{16} da_t^2 - \frac{1}{4} \frac{da_t^2}{\varepsilon} \right) M_s^{-2} \quad (116)$$

$$(y_e)^{wu} (y_e)^{vx} (y_e^\dagger)^{uv} \varepsilon_{il} \varepsilon_{jk} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} H_i H_j (H^\dagger)_k (L^\dagger)_l^{\dot{\alpha}_1, w} (e_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, x}$$

nD: 0, H: 2, H+: 1, Q+: 1, dC+: 1

term83

$$- \left(\frac{1}{16} d \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{32} da_t^2 - \frac{1}{8} \frac{da_t^2}{\varepsilon} \right) M_s^{-2} \quad (117)$$

$$(y_d)^{wu} (y_d)^{vx} (y_d^\dagger)^{uv} \varepsilon_{il} \varepsilon_{jk} \varepsilon_{cab} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} H_i H_j (H^\dagger)_k (Q^\dagger)_{l, ab}^{\dot{\alpha}_1, w} (d_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, x}$$

term85

$$\begin{aligned}
& - \left(\frac{1}{16} d \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{32} da_t^2 - \frac{1}{8} \frac{da_t^2}{\varepsilon} \right) M_s^{-2} \\
& (y_d)^{wu} (y_d)^{vx} (y_d^\dagger)^{uv} \varepsilon_{il} \varepsilon_{jk} \varepsilon_{cab} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} H_i H_j (H^\dagger)_k (Q^\dagger)_{l,ab}^{\dot{\alpha}_1,w} (d_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2,x}
\end{aligned} \tag{118}$$

term103

$$\begin{aligned}
& + \left(\frac{1}{16} d \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{32} da_t^2 - \frac{1}{8} \frac{da_t^2}{\varepsilon} \right) M_s^{-2} \\
& (y_u)^{wu} (y_u^\dagger)^{uv} (y_d)^{vx} \varepsilon_{ij} \varepsilon_{lk} \varepsilon_{cab} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} H_i H_j (H^\dagger)_k (Q^\dagger)_{l,ab}^{\dot{\alpha}_1,w} (d_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2,x}
\end{aligned} \tag{119}$$

term104

$$\begin{aligned}
& - \left(\frac{1}{16} d \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{32} da_t^2 - \frac{1}{8} \frac{da_t^2}{\varepsilon} \right) M_s^{-2} \\
& (y_u)^{wu} (y_u^\dagger)^{uv} (y_d)^{vx} \varepsilon_{ji} \varepsilon_{kl} \varepsilon_{cab} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} H_i H_j (H^\dagger)_k (Q^\dagger)_{l,ab}^{\dot{\alpha}_1,w} (d_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2,x}
\end{aligned} \tag{120}$$

nD: 0, H: 1, H+: 2, Q+: 1, uC+: 1

term94

$$\begin{aligned}
& + \left(\frac{1}{16} d \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{32} da_t^2 - \frac{1}{8} \frac{da_t^2}{\varepsilon} \right) M_s^{-2} \\
& (y_u)^{wu} (y_u)^{vx} (y_u^\dagger)^{uv} \varepsilon_{ij} \varepsilon_{lk} \varepsilon_{cab} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} H_i (H^\dagger)_j (H^\dagger)_k (Q^\dagger)_{l,ab}^{\dot{\alpha}_1,w} (u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2,x}
\end{aligned} \tag{121}$$

term95

$$\begin{aligned}
& + \left(\frac{1}{16} d \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{32} da_t^2 - \frac{1}{8} \frac{da_t^2}{\varepsilon} \right) M_s^{-2} \\
& (y_u)^{vx} (y_d)^{wu} (y_d^\dagger)^{uv} \varepsilon_{il} \varepsilon_{jk} \varepsilon_{cab} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} H_i (H^\dagger)_j (H^\dagger)_k (Q^\dagger)_{l,ab}^{\dot{\alpha}_1,w} (u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2,x}
\end{aligned} \tag{122}$$

term99

$$\begin{aligned}
& - \left(\frac{1}{16} d \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{32} da_t^2 - \frac{1}{8} \frac{da_t^2}{\varepsilon} \right) M_s^{-2} \\
& (y_u)^{wu} (y_u)^{vx} (y_u^\dagger)^{uv} \varepsilon_{ik} \varepsilon_{jl} \varepsilon_{cab} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} H_i (H^\dagger)_j (H^\dagger)_k (Q^\dagger)_{l,ab}^{\dot{\alpha}_1,w} (u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2,x}
\end{aligned} \tag{123}$$

term101

$$\begin{aligned}
& - \left(\frac{1}{16} d \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{32} da_t^2 - \frac{1}{8} \frac{da_t^2}{\varepsilon} \right) M_s^{-2} \\
& (y_u)^{vx} (y_d)^{wu} (y_d^\dagger)^{uv} \varepsilon_{il} \varepsilon_{jk} \varepsilon_{cab} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} H_i (H^\dagger)_j (H^\dagger)_k (Q^\dagger)_{l,ab}^{\dot{\alpha}_1,w} (u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2,x}
\end{aligned} \tag{124}$$

nD: 0, Q: 1, uC: 1, H: 2, H+: 1

term96

$$\begin{aligned}
& + \left(\frac{1}{16} d \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{32} da_t^2 - \frac{1}{8} \frac{da_t^2}{\varepsilon} \right) M_s^{-2} \\
& (y_u^\dagger)^{xu} (y_d)^{uv} (y_d^\dagger)^{vw} \varepsilon_{il} \varepsilon_{kj} \varepsilon_{abc} \varepsilon^{\alpha_1 \alpha_2} Q_{\alpha_1,i,a}^w (u_{\mathbb{C}})^x_{\alpha_2,bc} H_j H_k (H^\dagger)_l
\end{aligned} \tag{125}$$

term97

$$\begin{aligned}
& - \left(\frac{1}{16} d \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{32} da_t^2 - \frac{1}{8} \frac{da_t^2}{\varepsilon} \right) M_s^{-2} \\
& (y_u)^{uv} (y_u^\dagger)^{xu} (y_u^\dagger)^{vw} \varepsilon_{jl} \varepsilon_{ki} \varepsilon_{abc} \varepsilon^{\alpha_1 \alpha_2} Q_{\alpha_1,i,a}^w (u_{\mathbb{C}})^x_{\alpha_2,bc} H_j H_k (H^\dagger)_l
\end{aligned} \tag{126}$$

term98

$$\begin{aligned}
& - \left(\frac{1}{16} d \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{32} d a_t^2 - \frac{1}{8} \frac{d a_t^2}{\varepsilon} \right) M_s^{-2} \\
& (y_u)^{uv} (y_u^\dagger)^{xu} (y_u^\dagger)^{vw} \varepsilon_{ji} \varepsilon_{kl} \varepsilon_{abc} \varepsilon^{\alpha_1 \alpha_2} Q_{\alpha_1, i, a}^w (u_{\mathbb{C}})^x_{\alpha_2, bc} H_j H_k (H^\dagger)_l
\end{aligned} \tag{127}$$

term100

$$\begin{aligned}
& - \left(\frac{1}{16} d \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{32} d a_t^2 - \frac{1}{8} \frac{d a_t^2}{\varepsilon} \right) M_s^{-2} \\
& (y_u^\dagger)^{xu} (y_d)^{uv} (y_d^\dagger)^{vw} \varepsilon_{il} \varepsilon_{kj} \varepsilon_{abc} \varepsilon^{\alpha_1 \alpha_2} Q_{\alpha_1, i, a}^w (u_{\mathbb{C}})^x_{\alpha_2, bc} H_j H_k (H^\dagger)_l
\end{aligned} \tag{128}$$